SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Dhanush	**Reg. No.:**	1924260023
Date:	5/10/2026	**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**
   - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/frame work and provides	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing;

	strong technical justification.			report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

ONLINE EMERGENCY VEHICLE ROUTING SYSTEM

1. Abstract

The Online Emergency Vehicle Routing System is a web-based system designed to help emergency vehicles such as ambulances, fire engines, and rescue vehicles reach their destinations quickly and safely. The system identifies the emergency location and provides an efficient route to the nearest available emergency vehicle. It can help reduce travel time by considering road conditions, distance, and traffic information. The main objective is to improve emergency response time and provide better coordination between emergency vehicles and users.

2. Introduction

During emergencies, every second is important. Ambulances, fire engines, and other emergency vehicles may be delayed because of traffic congestion, unfamiliar routes, or lack of information about nearby vehicles. The Online Emergency Vehicle Routing System provides a digital solution for these problems.

The system allows users to report an emergency location and helps identify a suitable emergency vehicle. It then determines an efficient route from the vehicle's current location to the emergency location and finally to the required destination.

3. Problem Statement

In the existing emergency response system, emergency vehicles may take more time to reach the incident location because of traffic, incorrect route selection, and lack of real-time coordination.

Therefore, there is a need for an online system that can:

* Identify the emergency location.
* Locate available emergency vehicles.
* Select a suitable vehicle.
* Find an efficient route.
* Reduce emergency response time.
* Improve communication between users and emergency services.

4. Existing System

In the existing system, emergency vehicle requests are often handled through telephone calls or manual communication. Vehicle allocation and route selection may depend on the availability of operators.

Disadvantages

* Delay in vehicle allocation.
* Difficulty in identifying the nearest vehicle.
* Traffic may increase travel time.
* Manual communication can cause errors.
* Lack of centralized information.
* Route selection may not always be efficient.

5. Proposed System

The proposed Online Emergency Vehicle Routing System provides an online platform for managing emergency vehicle requests and routing.

When an emergency request is received, the system identifies the user's location and checks the availability of emergency vehicles. The nearest suitable vehicle can then be selected, and the system provides an efficient route to the emergency location.

Main Features

* Online emergency request.
* Location identification.
* Emergency vehicle tracking.
* Vehicle availability checking.
* Route calculation.
* Driver/user communication.
* Emergency status updates.
* Admin monitoring.

6. Objectives

The main objectives of the system are:

1. To provide quick emergency vehicle services.
2. To identify the nearest available emergency vehicle.
3. To calculate an efficient route.
4. To reduce emergency response time.
5. To improve coordination between users, drivers, and administrators.
6. To provide real-time status information.
7. To maintain emergency request and vehicle details.

7. Scope of the Project

The system can be used for different types of emergency services, including:

* Ambulance services.
* Fire and rescue services.
* Police/emergency response services.
* Disaster management.
* Road accident response.
* Medical emergency assistance.

The system can be expanded in the future to support real-time traffic data, GPS tracking, and automatic route optimization.

8. Modules

8.1 User Module

The user can:

* Register and log in.
* Submit an emergency request.
* Provide the emergency location.
* Select the type of emergency.
* View vehicle status.
* Track the assigned vehicle.

8.2 Emergency Request Module

This module collects emergency information such as:

* User details.
* Emergency type.
* Current location.
* Destination.
* Contact information.
* Emergency priority.

8.3 Vehicle Management Module

This module maintains information about emergency vehicles.

It includes:

* Vehicle number.
* Vehicle type.
* Driver details.
* Current location.
* Availability status.
* Vehicle status.

8.4 Routing Module

The routing module calculates an efficient route between the emergency vehicle and the emergency location.

It considers factors such as:

* Distance.
* Road connectivity.
* Available routes.
* Traffic conditions, if integrated.
* Destination location.

8.5 Driver Module

The driver can:

* Login to the system.
* Receive emergency requests.
* View the emergency location.
* View the suggested route.
* Update vehicle status.
* Mark the emergency request as completed.

8.6 Admin Module

The administrator manages the complete system.

The admin can:

* Manage users.
* Manage drivers.
* Manage vehicles.
* View emergency requests.
* Monitor vehicle status.
* Manage system information.
* Generate reports.

9. System Workflow

User → Emergency Request → Location Detection → Available Vehicle Search → Vehicle Selection → Route Calculation → Driver Notification → Vehicle Travels → Emergency Completed

10. Functional Requirements

The system should provide the following functions:

* User registration and login.
* Emergency request submission.
* Location detection.
* Vehicle availability management.
* Emergency vehicle assignment.
* Route calculation.
* Driver notification.
* Request status management.
* Admin management.
* Data storage and retrieval.

11. Non-Functional Requirements

The system should provide:

* Reliability: The system should work consistently during emergency situations.
* Performance: Requests should be processed quickly.

* Security: User and emergency information should be protected.
* Usability: The interface should be simple and easy to use.
* Scalability: The system should support more users and vehicles in the future.
* Availability: The system should be accessible whenever emergency services are required.

12. Advantages

* Reduces emergency response time.
* Helps identify nearby available vehicles.
* Provides efficient route information.
* Reduces manual work.
* Improves communication.
* Provides centralized emergency information.
* Helps administrators monitor emergency vehicles.
* Can support real-time vehicle tracking.

13. Applications

The system can be applied in:

* Hospitals.
* Ambulance services.
* Fire stations.
* Police departments.
* Emergency control centers.
* Disaster management organizations.
* Smart city emergency services.

14. Future Enhancements

Future versions of the system can include:

* Real-time GPS tracking.
* Real-time traffic information.
* AI-based route optimization.
* Automatic vehicle assignment.
* Voice-based emergency requests.
* Mobile application.
* Automatic accident detection.
* Integration with hospital systems.
* Live estimated time of arrival (ETA).
* Priority-based emergency routing.

15. Conclusion

The Online Emergency Vehicle Routing System provides an efficient digital solution for managing emergency vehicle services. By identifying emergency locations, selecting available vehicles, and providing suitable routes, the system can help improve emergency response and coordination. The system

can be further enhanced with GPS, real-time traffic information, and AI-based routing to provide more advanced emergency services.